

An evolutionary multi-objective approach to learn from positive and unlabeled data

Jianfeng Qiu^{a,c}, Xiaoqiang Cai^a, Xingyi Zhang^{a,b}, Fan Cheng^{a,*}, Shenzhi Yuan^a, Guanglong Fu^a

^a Key Laboratory of Intelligent Computing and Signal Processing of Ministry of Education, School of Computer Science and Technology, Anhui University, Hefei 230601, China

^b State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University, Shenyang 110819, China

^c Jiangsu Key Laboratory of Data Science and Smart Software, Jinling Institute of Technology, Nanjing 211169, China

ARTICLE INFO

Article history:

Received 11 April 2020

Received in revised form 9 August 2020

Accepted 30 November 2020

Available online 8 December 2020

Keywords:

PU learning

Multi-objective problem

Evolutionary algorithm

PU similarity

Elite label

ABSTRACT

Positive and unlabeled (PU) learning has attracted increasing interests in recent years. Despite that a number of PU learning algorithms have been proposed, most of them are subject to some assumptions about unlabeled sample distribution and objective functions, which makes them difficult to be adopted for real applications. To this end, in this paper, an evolutionary multi-objective approach, namely MOEA-PUL, is suggested for PU learning, whose aim is to build a PU classifier without any prior assumption for data distribution and objective functions. To be specific, the PU learning is formulated as a bi-objective optimization problem, where the true positive rate (TPR) and a new metric, termed unlabeled accuracy rate (UAR) are used as the two objectives. A multi-objective evolutionary algorithm is proposed to solve this bi-objective optimization problem, under the framework of NSGA-II, where a PU similarity based initialization strategy and an elite label based learning strategy are developed. Empirical studies on 12 datasets demonstrate the superiority of MOEA-PUL over the existing PU learning algorithms.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction



Positive and unlabeled (PU) learning has attracted much attention in different areas [1]. Different from the traditional supervised machine learning methods, in PU learning, only positive samples and a large number of unlabeled samples are available [2]. Therefore, it is difficult to train a binary classifier on the PU data using the supervised learning algorithms. To tackle the issue, a variety of algorithms have been proposed for PU learning based on different ideas of utilizing the unlabeled data. For example, some researchers focused on heuristically identifying several unlabeled samples as “reliable” negative samples for training the binary classifier [3–6]. Since only a small number of unlabeled samples are considered, the learnt classifiers are usually hard to achieve desirable performance. To this end, some researchers proposed to treat all the unlabeled data as negative samples with different weights [7,8]. Experimental results have demonstrated their superiority over the algorithms that only consider part of the unlabeled data. Different from the algorithms mentioned above,

some recent works for PU learning have been reported [9–12], where each unlabeled sample is viewed as positive and negative (PN). Then, the traditional supervised learning algorithms are adopted to build the PU classifiers, where a variety of loss functions have been used, such as hinge loss [10] and hat loss [9] etc. Empirical results have shown that it is effective to regard each unlabeled sample as positive and negative for training PU classifiers, which can achieve PU classifiers with better performance.

As analyzed above, many promising algorithms have been developed for PU learning in recent years. But, most of them are subject to the prior information on unlabeled data, e.g., the class prior in unlabeled data [10,12], the sample distribution in unlabeled data [9,11,13], which are often hard to be obtained in real applications. In addition, to utilize the traditional gradient based optimization algorithms, some restriction conditions (e.g., smooth and differentiable) on the loss functions are needed to be satisfied for finding the optimal PU classifier, such as in [9,10].

To this end, in this work, we adopt the multi-objective evolutionary algorithm (MOEA) to tackle the PU learning problem, which utilizes the advantage that does not require any prior information and restriction on optimization problem. Since MOEAs have shown the superiority to solve multi-objective problems (MOPs), different MOEAs have been proposed in which the concept of Pareto dominance is defined to find the non-dominated

* Corresponding author.

E-mail addresses: qiu Jianfeng@ahu.edu.cn (J. Qiu), saiqiang413@gmail.com (X. Cai), xyzhanghust@gmail.com (X. Zhang), chengfan@mail.ustc.edu.cn (F. Cheng), yuanshenzhi@outlook.com (S. Yuan), 1332937242@qq.com (G. Fu).

solutions and Pareto optimal set [14]. Among them, NSGA-II [14] is one of popular MOEAs in which the non-dominated sorting approach was proposed to partition the population into different fronts. In addition, the crowding distance and the elite strategy were designed to enhance the diversity and convergence of the population. The experimental results have demonstrated that NSGA-II can solve the multi-objective optimization problems well. Based on the work of NSGA-II, some variants of NSGA-II have been proposed to solve many-objective optimization problem. For example, Deb et al. [15] developed a reference-point based EA, named NSGA-III, to tackle with many-objective optimization problems. Furthermore, Seada et al. [16] proposed an integrated algorithm by considering more fine-grained factors, termed B-NSGA-III, to effectively solve many-objective optimization problems.

Despite NSGA-II and its variants have been widely used for solving different machine learning problems [17,18], to the best of our knowledge, there is no work on designing the MOEA for PU learning. To this end, in this paper, we propose an MOEA under the framework of NSGA-II for PU learning, which does not need any prior information on unlabeled samples and assumptions for objective functions. The main contributions of this paper are summarized as follows.

- (1) The PU learning is converted into a bi-objective optimization problem in which two objectives are designed to evaluate the performance of the PU classifier. More specifically, in the converted bi-objective problem, one objective is the widely used measure in labeled data, true positive rate (TPR) and the other is a new metric named unlabeled accuracy rate (UAR), which is developed to evaluate the quality of a set of candidate labels for unlabeled data.
- (2) A multi-objective evolutionary algorithm, namely MOEA-PUL is suggested to address the obtained bi-objective optimization problem. Due to the advantages of evolutionary algorithms, the suggested algorithm does not require any assumption on the distribution of the unlabeled data and restrictions on the objective functions. In the proposed algorithm, a PU similarity based population initialization strategy and an elite label based learning strategy are also designed to ensure that the proposed method to achieve better performance.
- (3) The effectiveness of the proposed MOEA-PUL is verified by comparing it with four state-of-the-art PU learning algorithms on 12 real-world datasets with different characteristics. Experimental results indicate the superior performance of the proposed algorithm in case that no prior information on unlabeled data is used.

The remainder of this paper is organized as follows. In Section 2, the preliminaries about PU learning and related work are presented. Section 3 presents the details of the proposed approach MOEA-PUL and analyzes the computational complexity of MOEA-PUL. Section 4 reports the experimental results by comparing it with four state-of-the-art PU learning algorithms on 12 real-world datasets. Finally, the conclusion and future work are discussed in Section 5.

2. Related work

2.1. PU learning

PU learning has been a hot research topic in machine learning community, which aims to build a binary classifier using only the positive and unlabeled data to predict the labels of the unseen samples [1]. In PU learning, besides a large number of unlabeled samples, only positive samples can be collected [2].

Taking the surveillance and monitoring system as an example, some behaviors are easily annotated as positive samples (normal behaviors). But, for a large number of behaviors, it is difficult to determine whether these behaviors are normal or abnormal, and hence treated as unlabeled samples. Since only having the positive samples, the traditional supervised (positive and negative) learning algorithms usually cannot be used for PU learning problems. Therefore, how to design an effective algorithm for PU learning is an important issue.

In this paper, we consider the binary classification PU problem. Given a dataset $D = \{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ is the i th sample with d components, y_i represents the label of x_i (positive or unlabeled) and n is the number of samples in D . Obviously, the samples in D can be divided into two subsets, namely, a set of positive samples and a set of unlabeled samples. The PU learning problem is to construct a classifier function $C(x)$ with good generalization performance, where C denotes the PU classifier trained on the PU data, x represents the unseen samples.

2.2. Related work on PU learning

Since the PU data has widely existed in many real applications, many efforts have been devoted for PU learning and a large number of algorithms developed for PU learning have been proposed [3,4,7–12,19,20], which can be roughly divided into three groups as follows.

The first group of algorithms is to heuristically identify the part of unlabeled samples as negative ones, which are further combined with the original positive samples to build PU classifiers. Li et al. [3] proposed a PU learning algorithm by extracting some “reliable” negative samples from the unlabeled data, which is together with the positive samples for training PU classifier by a standard supervised learning algorithm. Xiao et al. [19] further extended the idea and proposed a similarity based PU learning algorithm to identify the labels of the remaining unlabeled data, which cannot be identified as “reliable” negative samples, for obtaining PU classifier with better performance. The similar idea was also utilized for solving PU learning problem in [4–6]. However, one disadvantage of the first group of algorithms is that due to only consider part of unlabeled samples, the performance of the obtained PU classifiers are not very promising [9].

The second group of algorithms treats all the unlabeled data as negative samples with noise and thus converts PU learning into a noisy label learning problem. Lee et al. [7] developed a weighted logistic regression model to deal with the converted noisy label learning problem, where the regularization parameters for the weights were needed to be estimated according to the sample distribution of unlabeled data. Experimental results confirmed the superiority of the algorithm over the algorithms that only consider part of unlabeled data. To further address the false negative label noise, Shi et al. [20] proposed a novel PU learning algorithm to decompose the empirical loss on noisy negative data for estimating the centroid of noisy negative examples. Recently, Northcutt et al. [21] proposed a novel algorithm, named CL, for PU learning in which the unlabeled samples were regarded as the noisy negative samples and the joint distribution between the noisy labels and the true latent labels were estimated to identify the reliable negative samples with the aim to train the PU classifier. Similar works belonging to this group are included in [22–24].

Different from the above two groups of algorithms that only assuming the unlabeled data as negative samples, the third group of algorithms is to regard each unlabeled sample as positive and negative class. A representative work, named uPU, was proposed by Plessis et al. [12], where PU learning was transformed into a cost-sensitive PN learning problem. To address the converted

cost-sensitive learning problem, the class prior was needed to be estimated by different approaches [8,25], since the true class prior was usually unknown in practice. Experimental results showed that the algorithm can achieve a better PU classifier than the algorithms that treating all unlabeled data as negative. Based on the work developed in [12], Plessis et al. [10] further adopted different loss functions on positive and unlabeled data to find the global optimal solution. Recently, Zhang et al. [11] developed a novel PU learning approach, termed PULD, where each unlabeled sample was ambiguously labeled as positive and negative class and PU learning was transformed into a partial label learning problem [26]. In PULD, the sample distribution in unlabeled data needed to be known in advance, which was used to build the PU classifier. Experiments demonstrated the superiority of the algorithm over the state-of-the-art PU learning algorithms.

We note that besides the algorithms mentioned above, there also exist some algorithms that regard PU learning as a special case of PNU (positive, negative and unlabeled) learning. For example, Sakai et al. [27] incorporated negative data into PU classification and proposed a novel semi-supervised learning approach. Experimental analysis demonstrated that the suggested PNU classification was effective in terms of both accuracy and efficiency.

From the above analysis, we can find that a number of promising algorithms have been developed for PU learning, but most of them are subject to some prior knowledge, e.g., the sample distribution on unlabeled data [9,11,13], or the prior information over classes in unlabeled samples [10,12], etc, which are often hard to be available in advance for real-life applications. In addition, there are also some restrictions (e.g., smooth and differentiable) on the adopted loss functions to employ the traditional gradient based optimization methods [9,10], which makes it very challenging to design suitable loss functions for PU learning. Since MOEAs do not require prior knowledge to the optimized problems and have no any restriction on the objectives to be optimized, they have widely been used to solve the MOPs with discontinuous, nonconvex, discrete optimization objectives. Therefore, in this paper, we suggest an evolutionary multi-objective approach under the framework of NSGA-II for PU learning, which can be used to optimize two or three objectives simultaneously. The details of the proposed evolutionary multi-objective approach will be presented in Section 3.

3. The proposed algorithm MOEA-PUL

In this section, we elaborate the details of the proposed evolutionary multi-objective approach MOEA-PUL. First, we present the bi-objective optimization problem formulated in MOEA-PUL for PU learning. Then, a multi-objective evolutionary algorithm is given to tackle the bi-objective optimization problem, followed by two important strategies developed in MOEA-PUL for achieving the PU classifier with high quality. To help readers follow this paper conveniently, the main abbreviation descriptions used in this paper are firstly listed in Table 1.

3.1. Formulation of the bi-objective optimization problem for PU learning



In existing PU learning algorithms, the prior knowledge about unlabeled data often needs to be known in advance so as to construct the PU classifier, which is not suitable in real-life applications. To this end, in this paper, we formulate the PU learning into the following bi-objective optimization problem without

Table 1

The main abbreviation descriptions used in this paper.

Abbreviation	Description
PU	The positive and unlabeled
PN	The positive and negative
TPR	The true positive rate
FPR	The false positive rate
UAR	The unlabeled accuracy rate
MOP	The multi-objective problem
MOEA	The multi-objective evolutionary algorithm
PNU	The positive, negative and unlabeled
$I(\cdot)$	The indicator function
$C(\cdot)$	The classification function
X_p	The set of positive samples
X_U	The set of unlabeled samples
$f_{TPR}(X_p)$	The true positive rate on the X_p
$f_{UAR}(X_U)$	The unlabeled accuracy rate on the X_U
p_c	The crossover probability

utilizing any prior knowledge:

$$\begin{cases} f_{TPR}(X_p) = \sum_{t=1}^{|X_p|} I(\text{sign}(C(x_p^t)) = 1) / |X_p| \\ f_{UAR}(X_U) = \sum_{i=1}^{|X_{U-}|} \sum_{j=i+1}^{|X_{U+}|} I(C(x_{U-}^i) > C(x_{U+}^j)) / (|X_{U-}| \cdot |X_{U+}|) \end{cases} \quad (1)$$

where X_p and X_U denote the set of positive samples and the set of unlabeled sample in PU data, X_{U+} and X_{U-} are the sets of positive and negative samples identified from unlabeled data X_U , satisfying $X_U = X_{U+} \cup X_{U-}$. The x_p^t is the t th positive sample in X_p , x_{U-}^i and x_{U+}^j respectively denote the i th and j th negative and positive samples identified from X_U , $|\cdot|$ represents the number of samples for a given set and $C(\cdot)$ is the classification function learnt from the PU data, which is used to calculate the prediction score for a given sample. $\text{sign} : \mathbb{R} \rightarrow \{-1, 1\}$ is the map function used to achieve the labels of samples according to the prediction scores, where $\text{sign}(\cdot) = 1$ (positive class) if the prediction score is greater than the given threshold used in classification model, and $\text{sign}(\cdot) = -1$ (negative class) otherwise. $I(\cdot)$ is the indicator function, $I(\cdot) = 1$ if the argument is true, and $I(\cdot) = 0$ otherwise.

The first objective $f_{TPR}(X_p)$ is a widely adopted metric [28] used to measure the accuracy of the PU classifier on the positive samples, i.e. the percentage of positive samples correctly classified by the classifier to all positive samples in X_p . However, TPR is difficult to be directly used as an objective function, since it is not smooth and differentiable. To this end, different surrogate functions with smooth and differentiable have been designed as the objective functions [9,10] in the most of existing PU learning algorithms. Different from them, in this paper, we utilize the advantages of evolution algorithms where no any assumption (e.g., smooth and differentiable) about the objective function is needed so as to directly adopt the metric TPR as an objective function to be optimized. Maximizing $f_{TPR}(X_p)$ is expected to achieve the PU classifier with high accuracy on the positive samples.

The second objective $f_{UAR}(X_U)$ represents the percentage that the prediction score of identified negative sample in unlabeled data is greater than that of the positive one identified from unlabeled data to all pairwise (positive and negative) samples, which is inspired by the following intuition. For each negative sample identified from unlabeled data, the number of positive samples in unlabeled data whose prediction scores are smaller than that of the negative sample should be minimal. Since the suggested metric UAR does not need the prior information about unlabeled data, we formulate the metric as the second objective to evaluate the performance of PU classifier on unlabeled data. Minimizing $f_{UAR}(X_U)$ is to obtain the PU classifier with high performance on

The Unlabeled sample set X_U	x_u^1	x_u^2	x_u^3	x_u^4	x_u^5	x_u^6	x_u^7	x_u^8
The identified label	N	P	N	P	P	P	N	P
The prediction score $C(\cdot)$	0.88	0.50	0.71	0.43	0.64	0.45	0.59	0.78

$$X_{U^+} = \{x_u^2, x_u^4, x_u^5, x_u^6, x_u^8\} \quad X_{U^-} = \{x_u^1, x_u^3, x_u^7\} \quad (a)$$

$$\begin{aligned} \sum_{x_u^j \in X_{U^+}} I(C(x_u^1) > C(x_u^j)) &= 5 \\ \sum_{x_u^j \in X_{U^+}} I(C(x_u^3) > C(x_u^j)) &= 4 \\ \sum_{x_u^j \in X_{U^+}} I(C(x_u^7) > C(x_u^j)) &= 3 \end{aligned} \quad (b)$$

$$f_{UAR}(X_U) = \frac{5 + 4 + 3}{3 * 5} = 0.8 \quad (c)$$

Fig. 1. An example to illustrate the process of calculating the second objective $f_{UAR}(X_U)$. 'P' and 'N' represent the identified positive and negative label respectively.

unlabeled data. In conclusion, $f_{TPR}(X_P)$ is to measure the performance of the PU classifier on the Positive samples while $f_{UAR}(X_U)$ is proposed to evaluate the classifier on Unlabeled samples. By simultaneously optimizing the suggested two objective functions, we can achieve a PU classifier with promising performance.

Fig. 1 gives an example to illustrate the process of calculating the second objective $f_{UAR}(X_U)$. In this example, there exist eight unlabeled samples, where $\{x_{U-}^1, x_{U-}^3, x_{U-}^7\}$ are identified as negative samples and the remaining unlabeled samples are regarded as positive ones (see Fig. 1(a)). First, for each identified negative sample (e.g., x_{U-}^1), the number of positive samples whose prediction scores are smaller than $C(x_{U-}^1)$ is calculated. Similarly, for the other negative samples, the numbers of the corresponding positive samples are computed (see Fig. 1(b)). Second, the percentage that the sum of the number of positive samples (5 + 4 + 3) to the number of the all pairwise (positive and negative) samples (3 * 5) is calculated as the value of the second objective $f_{UAR}(X_U)$ (see Fig. 1(c)).

3.2. General framework of MOEA-PUL

Due to the fact that the suggested optimization objectives are not smooth, most of the traditional gradient-based optimization techniques are not suitable to tackle them. To this end, an evolutionary computation based multi-objective algorithm, named MOEA-PUL, is developed for the bi-objective optimization problem to obtain a PU classifier with high quality. In this section, we present the framework of the proposed MOEA-PUL.

Similar to the third group of PU learning algorithms, in this paper, we also focus on how to identify accurate positive and negative samples from unlabeled data to build PU classifier [10–12]. Different from the methods that identifying positive and negative samples by various assumptions, we adopt the evolutionary mechanism to distinguish positive and negative ones from unlabeled data and obtain an optimal PU classifier. Hence, the binary encoding is adopted to represent the identified positive or negative labels of unlabeled samples. More specifically, an individual $ind = (u_1, \dots, u_i, \dots, u_n)$, is adopted to encode the labels of unlabeled data, where n is the number of unlabeled samples. The u_i denotes the identified label of the i th unlabeled sample, it is regarded as positive if $u_i = 1$, and negative $u_i = 0$ otherwise. Taking the individual $ind = (0, 1, 0, 1, 1, 1, 0, 1)$ as an example, the eight unlabeled samples are divided into two sets according to the identified labels, the unlabeled samples $x_{U-}^2, x_{U-}^4, x_{U-}^5, x_{U-}^6$ and x_{U-}^8 are identified as positive samples (denoted as 1), x_{U-}^1, x_{U-}^3 and x_{U-}^7 are regarded as negative samples (denoted as 0).

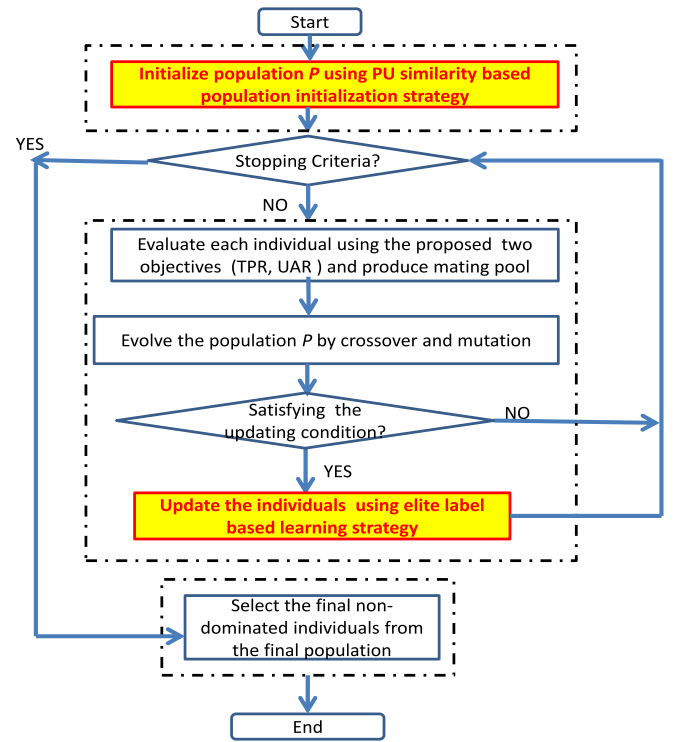


Fig. 2. A flowchart of the proposed MOEA-PUL.

Based on the employed binary representation, the proposed MOEA-PUL adopts a framework similar to NSGA-II [14]. Fig. 2 gives a flowchart of the proposed MOEA-PUL to illustrate the process of build PU classifiers by evolutionary multi-objective approach. First of all, a population P of size N is initialized by using the proposed PU similarity based initialization strategy. Second, for each individual, the two objective function values are calculated using the proposed two objectives TPR and UAR . Then, a binary tournament mating selection is adopted to produce the mating pool. Specifically, two individuals are randomly selected from the population, if one individual dominates the other individual, then the former individual is selected. If the two individuals are nondominated with each other, then the algorithm will check their crowding distances. The individual with the larger crowding distance wins the tournament. If both individuals have an equal crowding distance, then one of them

Algorithm 1: Framework of the proposed MOEA-PUL

Require: X_U : unlabeled dataset; X_P : positive dataset; N : population size; p_c : crossover probability; p_m : mutation probability; G : the iteration starting to apply the elite label based learning strategy;

Ensure: The final set of non-dominated solutions PS

Step1: Population initialization.

- 1: $P \leftarrow \text{PuSimilarityInitialization}(X_U, X_P, N)$; // Algorithm 2
- //For each $p_i, i = (1, \dots, N)$, a PU classification function C_i is trained to calculate the suggested two objectives.
- 2: $g = 1$;
- Step2: Population evolution.**
- 3: **while** termination criterion not fulfilled **do**
- 4: $P' \leftarrow \text{MatingSelection}(P)$;
- // $2 * N$ individuals are selected from P as the parents using the binary tournament selection.
- 5: $O \leftarrow \text{Variation}(P', N)$;
- // For each o_j in $O, j = (1, \dots, N)$, a PU classification function C_j is trained to calculate the suggested two objectives.
- 6: **if** $g \geq G$ **then**
- 7: $P \leftarrow \text{EliteBasedLearningStrategy}(|X_U|, P)$; // Algorithm 3.
- 8: **end if**
- 9: $g = g + 1$;
- 10: $P \leftarrow \text{EnvironmentalSelection}(P \cup O, N)$;
- 11: **end while**
- Step3: The final solution selection.**
- 12: $PS \leftarrow \text{SelectNonDominatedsets}(P)$;
- //Select the final non-dominated individuals from the final population P .

will be randomly selected. After mating selection, N offsprings are generated by the crossover and mutation operations used in NSGA-II. To be specific, two individual p_1 and p_2 are randomly selected from the mating pool, and then the uniform crossover operator is adopted to generate two offsprings. For each dimension, e.g., the d th dimension, a real number $\text{rand}() \in (0, 1)$ is generated randomly, and if $\text{rand}() < p_c$ (p_c is the crossover probability), $p_{1,d}$ and $p_{2,d}$ are swapped to generate new offsprings. To enhance the performance of the proposed algorithm, an elite label based learning strategy is also developed to guide the population evolution. Afterwards, N individuals with better nondominated front number and crowding distance are selected from the combination of the population P and its offspring O to be survived to the next generation. The second step repeats until the maximum number of generations is reached. Finally, the non-dominated individuals in the final population are selected as the output.

The whole procedure of the proposed MOEA-PUL is illustrated in Algorithm 1, from which we can find that two important strategies, namely the PU similarity based initialization strategy (see line 1 in Algorithm 1) and the elite label based learning strategy (see line 7 in Algorithm 1). With these two strategies, MOEA-PUL can obtain the PU classifier with high quality. In the following part of this section, we will elaborate them in detail.

3.3. PU similarity based population initialization

Random initialization is a widely used strategy in MOEAs for creating an initial population. However, the randomly generated solutions may focus on some parts of the objective space while the other parts are difficult to be searched, which makes the random initialization hard to find the promising solutions [29]. To tackle this issue, we propose a PU similarity based population initialization strategy in MOEA-PUL to obtain the initial population with diversity.

The basic idea of the suggested initialization strategy is that for each unlabeled sample, a smaller sum of the distances between it and all positive samples indicates that it has a higher

Algorithm 2: PU Similarity Initialization(X_U, X_P, N)

Require: X_U : unlabeled dataset $X_U = [x_u^1, x_u^2, \dots, x_u^{|X_U|}]$; X_P : positive dataset $X_P = [x_p^1, x_p^2, \dots, x_p^{|X_P|}]$; N : population size;

Ensure: Initial population: $P = [p_1, p_2, \dots, p_N]$;

Step1: Calculate the PU similarity.

- 1: **for** $i = 1$ to $|X_U|$ **do**
- 2: $\text{PuDistance}[i] = \sum_{j=1}^{|X_P|} \text{Distance}(x_u^i, x_p^j)$;
- 3: **end for**
- 4: **for** $i = 1$ to $|X_U|$ **do**
- 5: $\text{PuSimilarity}[i] = \frac{\text{PuDistance}[i]}{\max_{m \in \{1, \dots, |X_U|\}} \text{PuDistance}[m]}$;
- 6: **end for**
- Step2: Initialize the population by the calculated PU similarity.**
- 7: **for** $k = 1$ to N **do**
- 8: $p_k = \text{zeros}(1, |X_U|)$; $m = 0$;
- 9: **while** $m \leq |\text{rand}() \times |X_U||$ **do**
- 10: randomly choose two unlabeled samples x_u^a and x_u^b from X_U ;
- 11: **if** $\text{PuSimilarity}[a] < \text{PuSimilarity}[b]$ **then**
- 12: $p_{k,a} = 1$;
- 13: **end if**
- 14: **if** $\text{PuSimilarity}[a] > \text{PuSimilarity}[b]$ **then**
- 15: $p_{k,b} = 1$;
- 16: **end if**
- 17: **if** $\text{PuSimilarity}[a] == \text{PuSimilarity}[b]$ **then**
- 18: **if** $\text{rand}() < 0.5$ **then**
- 19: $p_{k,a} = 1$;
- 20: **else**
- 21: $p_{k,b} = 1$;
- 22: **end if**
- 23: **end if**
- 24: $m = m + 1$;
- 25: **end while**
- 26: **end for**

similarity with them and is more likely to be identified as positive sample. To avoid the possible bias, a diversity maintain method is proposed in the PU similarity based population initialization strategy, to generate the promising initial population. More specifically, a random real number between 0 and 1 is firstly generated and the labels of unlabeled samples are determined by comparing it with the corresponding PU similarity. Algorithm 2 presents the proposed population initialization strategy of MOEA-PUL, which consists of the following two steps, i.e., calculating the PU similarity and initializing the population by the PU similarity.

The first step is to calculate the similarity between each unlabeled sample and all positive samples. More specifically, in an unlabeled sample set $X_U = \{x_u^1, \dots, x_u^i, \dots, x_u^{|X_U|}\}$, for each unlabeled sample $x_u^i \in X_U$, the sum of Euclidean distance between x_u^i and all positive samples is calculated by using Formula (2). Then, the normalized Euclidean distance of the unlabeled sample x_u^i is regarded as the similarity of x_u^i with all positive samples, called PU similarity of the i th unlabeled sample (in Formula (3)).

$$\text{PuDistance}_i = \sum_{j=1}^{|X_P|} \text{Distance}(x_u^i, x_p^j), i \in \{1, \dots, |X_U|\} \quad (2)$$

$$\text{PuSimilarity}_i = \frac{\text{PuDistance}_i}{\max_{m \in \{1, \dots, |X_U|\}} \text{PuDistance}_m} \quad (3)$$

where $\text{Distance}(x_u^i, x_p^j)$ is the Euclidean distance between the unlabeled sample x_u^i and the positive sample x_p^j , PuSimilarity_i represents the similarity of the unlabeled sample x_u^i with the positive samples, $|X_U|$ and $|X_P|$ are the numbers of unlabeled

samples and positive samples, respectively. The PU similarities of unlabeled samples will be used to generate the initial population, where the elements in each individual are determined according to the calculated PU similarity.

The second step is to generate a population P with N individuals by using the calculated PU similarity. To be specific, for the k th individual $p_k = (p_{k,1}, \dots, p_{k,i}, \dots, p_{k,|X_U|})$ in P , two unlabeled samples x_u^a and x_u^b are randomly selected. If $PuSimilarity_a < PuSimilarity_b$ (x_u^a has higher PU similarity than that of x_u^b), then the a th element in p_k is set to 1, i.e., $p_{k,a} = 1$, otherwise $p_{k,b} = 1$. If $PuSimilarity_a$ equals to $PuSimilarity_b$, then $p_{k,a}$ or $p_{k,b}$ will be randomly selected and assigned to 1. The process is repeated until the number of the positive samples in p_k reaches the value of $\lfloor rand() \times |X_U| \rfloor$, where $rand()$ denotes a uniformly distributed random number between (0, 1).

In Fig. 3, we take the individual p_k as an example to illustrate the suggested PU similarity based initialization process. The PU dataset consists of five positive samples (blue circle) and eight unlabeled samples (red circle), which are shown in Fig. 3(a). First, the PU similarities between each unlabeled sample and the five positive samples are calculated and shown in Fig. 3(b). Second, the PU similarity based initialization process is illustrated in Fig. 3(c). More specifically, two unlabeled samples x_u^1 and x_u^6 are selected randomly and then compare the PU similarity between them. The 6th element in p_k is assigned to 1, since the 6th unlabeled sample has higher similarity with positive samples than the 1st unlabeled sample, i.e., $PuSimilarity_6 < PuSimilarity_1$. Similarly, the 2nd, 4th, 5th and 8th elements in p_k are assigned to 1 until the number of positive samples reaches the generated random number which is 5 in this example. Fig. 3(d) gives the generated individual p_k , where the remaining elements in p_k , i.e., the 1st, 3rd and 7th elements, are assigned to 0.

3.4. Elite label based learning strategy

In MOEA-PUL, the crossover and mutation operations used in NSGA-II are employed to generate the new offsprings. To further enhance the effectiveness of search, an elite label based learning strategy is proposed. The basic idea is that in non-dominated individuals, for the same unlabeled sample, if all individuals assign 1 (or 0) to it, the corresponding label is treated as the elite label. These obtained elite labels are utilized to effectively guide the inferior individuals evolution. Algorithm 3 presents the suggested strategy in MOEA-PUL, which includes two steps: (1) the identification of the elite labels (Algorithm 4) and (2) the learning process guided by the obtained elite labels (Algorithm 5).

Algorithm 3: EliteBasedLearningStrategy($|X_U|$, P)

Require: $|X_U|$: The number of unlabeled samples, P : Population;
Ensure: P' : Updated population;
 1: $EliteLabelSet \leftarrow GeneratedEliteLabelSet(|X_U|, P)$ // Algorithm 4.
 2: $P' \leftarrow GuidedByEliteLabel(EliteLabelSet, P)$ // Algorithm 5.

At the first step, for each unlabeled samples x_u^i , a real number $temp_i$ is calculated by using the non-dominated individuals at the current generation, which is defined as follows.

$$temp_i = \frac{\sum_{m=1}^{|EliteSet|} EliteSet_{m,i}}{|EliteSet|} \quad (4)$$

where $EliteSet_{m,i}$ is the label value of the i th unlabeled sample (1 or 0) in the m th non-dominated individual and $|EliteSet|$ denotes the number of the non-dominated individuals at the current

Algorithm 4: GeneratedEliteLabelSet($|X_U|$, P)

Require: $|X_U|$: The number of unlabeled samples, P : Population;
Ensure: $EliteLabelSet$: The elite labels of the unlabeled samples;
 1: $EliteLabelSet = (-1) * ones(1, |X_U|)$;
 //Initialize the elite labels of the unlabeled samples as -1;
 2: $EliteSet \leftarrow SelectNonDominatedsets(P)$;
 // Select non-dominated individuals from the population P .
 3: **for** $i = 1$ to $|X_U|$ **do**
 $\sum_{m=1}^{|EliteSet|} EliteSet_{m,i}$
 4: $temp = \frac{\sum_{m=1}^{|EliteSet|} EliteSet_{m,i}}{|EliteSet|}$;
 5: **if** $temp == 0$ **then**
 6: $EliteLabelSet[i] = 0$;
 7: **end if**
 8: **if** $temp == 1$ **then**
 9: $EliteLabelSet[i] = 1$;
 10: **end if**
 11: **end for**

generation. The elite label of the i th unlabeled samples x_u^i is formulated as follows.

$$EliteLabelSet_i = \begin{cases} 1 & temp_i == 1 \\ 0 & temp_i == 0 \end{cases} \quad (5)$$

where $EliteLabelSet_i$ is the elite label of the unlabeled sample x_u^i . $EliteLabelSet_i = 1$ ($EliteLabelSet_i = 0$) represents that all non-dominated individuals regard the i th unlabeled sample as positive (negative). By using the Formula (4) and (5), we can obtain the elite labels of all unlabeled samples from the non-dominated individuals, which are used to update the individuals in population.

Algorithm 5: GuidedByEliteLabel($EliteLabelSet$, P)

Require: $EliteLabelSet$: The elite labels of the unlabeled samples, P : Population;
Ensure: P' : Updated population;
 1: $IndexP \leftarrow find(EliteLabelSet == 1)$;
 $IndexN \leftarrow find(EliteLabelSet == 0)$;
 2: $Len_P = length(IndexP)$; $Len_N = length(IndexN)$;
 3: $[frontnum, crowdist] \leftarrow NondominatedSort(P)$;
 //Obtain the non-dominated front number and crowding distance of each individual by non-dominated sorting.
 4: $n = 0$;
 5: **while** $n \leq \frac{|P|}{2}$ **do**
 6: randomly choose two individuals p_a and p_b from P ;
 7: **if** $frontnum(p_a) < frontnum(p_b)$ **then**
 8: $p_{b, \{IndexP\}} = ones(1, Len_P)$;
 9: $p_{b, \{IndexN\}} = zeros(1, Len_N)$;
 10: **end if**
 11: **if** $frontnum(p_a) > frontnum(p_b)$ **then**
 12: $p_{a, \{IndexP\}} = ones(1, Len_P)$;
 13: $p_{a, \{IndexN\}} = zeros(1, Len_N)$;
 14: **end if**
 15: **if** $frontnum(p_a) == frontnum(p_b)$ **then**
 16: **if** $crowdist(p_b) > crowdist(p_a)$ **then**
 17: $p_{a, \{IndexP\}} = ones(1, Len_P)$;
 18: $p_{a, \{IndexN\}} = zeros(1, Len_N)$;
 19: **else**
 20: $p_{b, \{IndexP\}} = ones(1, Len_P)$;
 21: $p_{b, \{IndexN\}} = zeros(1, Len_N)$;
 22: **end if**
 23: **end if**
 24: $n = n + 1$;
 25: **end while**
 26: $P' \leftarrow P$;

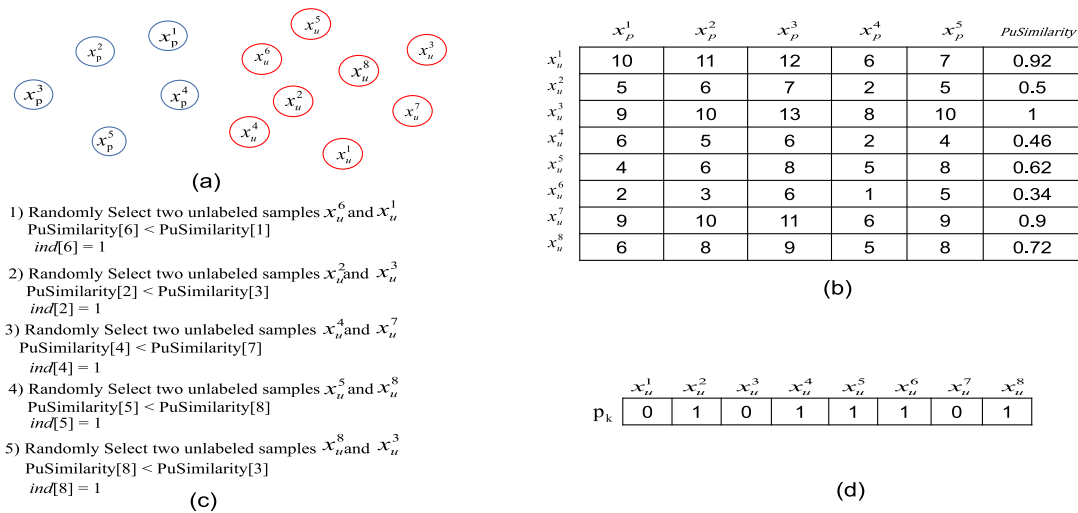


Fig. 3. An example to illustrate the PU similarity based initialization strategy.

At the second step, with the obtained elite labels of unlabeled samples, an elite label based learning strategy is performed to update the candidate individuals for improving the quality of the population. Specifically, two individuals are randomly selected from the population P . One of them is selected as the individual for updating according to the non-dominated front number and crowding distance of them [14]. The k th individual $p_k = (p_{k,1}, \dots, p_{k,i}, \dots, p_{k,|X_U|})$ is assumed as the selected individual for updating. For each element in p_k , e.g., $p_{k,i}$, if its elite label EliteLabelSet_i equals to 1 or 0, the i th element in p_k associated with the i th unlabeled sample is replaced with the elite label EliteLabelSet_i .

Fig. 4 presents an example to illustrate the elite label based learning strategy. There exist six non-dominated individuals and each one represents a set of PN labels for the eight unlabeled samples, EliteLabelSet is the set of the elite labels obtained from the six non-dominated individuals. Since the elite labels of $\{x_u^1, x_u^3, x_u^8\}$ are '1', the 1st, 3rd, 8th elements in the individual p_k are replaced with 1. Similarly, the 5th element in p_k is changed to 0, i.e. $p'_k = (1, 1, 1, 0, 0, 0, 0, 1)$.

3.5. Computational complexity analysis

In this section, an upper bound of the computational efforts and time of MOEA-PUL is provided as follows. Assuming that $|X_P|$ and $|X_U|$ denote the number of the positive samples and the unlabeled samples in the dataset, respectively. N is the population size and d represents the number of the unlabeled samples. According to Algorithm 1, MOEA-PUL includes three main steps, that is (1) population initialization; (2) population evolution and (3) the final solution selection. In the first step, the complexity of population initialization is $O(|X_P| \cdot |X_U|)$ for calculating the distances between the positive and unlabeled samples. In the second step, the complexity of population evolution includes the offspring generation and the environmental selection, which is denoted by $O(Nd)$ and $O(N \log(N))$, respectively. Finally, the non-dominated sorting is performed to select the final nondominated individuals, which needs a runtime of $O(N^2)$ in the worst case. Therefore, the computational efforts and time of one generation of MOEA-PUL is $O((|X_P| \cdot |X_U| + Nd + N \log(N) + N^2))$, which equals to $O(|X_P| \cdot |X_U|)$.

Table 2

12 datasets with different characteristic.

Dataset	#Samples	#Features	Num of P	Num of N
Sonar	208	60	97	111
Heart	270	13	120	150
Ionosphere	351	34	225	126
Wdbc	569	30	357	212
Balance	625	4	288	337
Breast	683	10	239	444
Transfusion	748	4	178	570
Pima	768	8	268	500
Diabetes	768	8	500	268
Vehicle	846	18	212	634
German	1000	24	300	700
Usps	9298	256	1553	7745

4. Experimental results and analysis

In this section, we verify the effectiveness of the proposed MOEA-PUL on solving PU learning problem by comparing it with four representative PU learning algorithms. To be specific, we first present the experimental settings. Then, the empirical results of MOEA-PUL and four baselines are reported. Finally, the effectiveness of the two suggested strategies and the sensitivity of parameter G are discussed.

4.1. Experimental setting

4.1.1. Datasets

In our experiments, 12 real-world binary datasets with different characteristics are adopted to evaluate the performance of the algorithms. These datasets are available in UCI machine learning repository¹ and the detailed characteristics are summarized in Table 2.

In this table, '#Samples' and '#Features' represent the numbers of samples and features, respectively. The numbers of positive sample and negative ones of each dataset are denoted as 'Num of P' and 'Num of N'. For each dataset, the widely adopted data preprocessing method is used to generate the corresponding PU dataset [9,11]. Specifically, we randomly select $\mu\%$ (i.e. $\mu \in \{20, 30, 40\}$) positive samples, together with all negative samples to compose the unlabeled data, and leave the rest positive samples as positive data. Under each μ , five-fold cross validation is

¹ <https://archive.ics.uci.edu/ml/>.

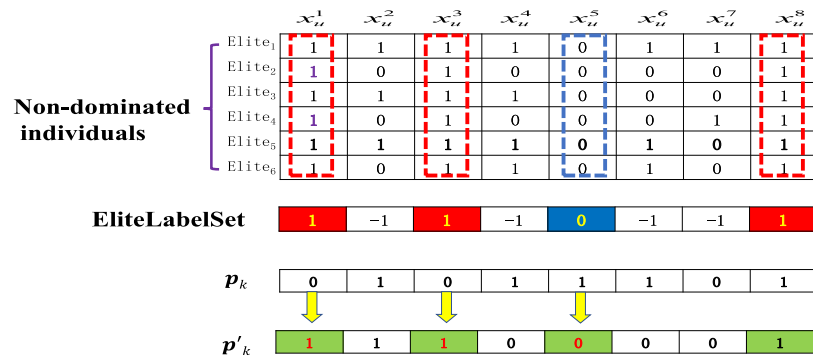


Fig. 4. An illustrative example of the elite label set based learning strategy, “-1” represents that for the unlabeled sample, some non-dominated individuals regard it as positive while the other ones treat it as negative.

employed, where four folds are used for training the PU classifier and the remaining fold for testing. The average results over the five independent implementations are reported.

4.1.2. Compared algorithms

To verify the effectiveness of the proposed MOEA-PUL, MOEA-PUL is compared with four state-of-the-art algorithms for PU learning, namely, CL [21], uPU [12], PULD [11] and PNU [27]. The CL is the approach that regards all the unlabeled samples as negative ones with noise and build PU classifier by identifying the confident negative ones from them. The uPU and PULD are the methods that treat the unlabeled samples as positive and negative, which have been demonstrated to achieve better performance in PU learning. Among them, the uPU is a classical PU learning algorithm which transforms PU learning into a cost-sensitive PN learning. In uPU, since the true class prior is unknown in real applications, it usually needs to be estimated for solving the cost-sensitive PN learning problem. The PULD is a recently proposed algorithm for PU learning problem by converting it to a partial label learning problem, where each unlabeled sample is ambiguously labeled as positive and negative. A margin based disambiguation strategy is suggested to identify the most likely label from them. In PULD, the sample distribution on unlabeled data needs to be known in advance for achieving better performance. To make the experiments more convincing, a general semi-supervised learning algorithm, named PNU, is also employed as a comparison algorithm, which can address not only for PNU learning but also for PU learning (setting the number of negative samples as 0). It should be noted that the four compared algorithms are single-objective based PU learning algorithms while the proposed MOEA-PUL is a multi-objective based algorithm for PU learning. For the sake of fairness, in the following experiments, we choose the one with the maximum accuracy on testing data from the results of MOEA-PUL as the final output.

For fair comparison, the recommended parameters values for the comparison algorithms are adopted, which are suggested by the authors in their original paper [11,12,21,27]. In MOEA-PUL, the population size of N is set to 100, the maximum number of generation is fixed to $maxgene = 100$, the crossover probability p_c is set to 0.9, and the mutation probability p_m is set to $\frac{1}{d}$ (d is the size of unlabeled dataset). The parameter G is set as 50, which represents the iteration starting to apply the elite label based learning strategy. In addition, we employ t-test with a significance level of 0.05 to investigate whether the proposed MOEA-PUL is significantly better than the baselines. All the algorithms are implemented on a computer equipped with an Intel i5-6500 processor with 3.2 GHz and 8 GB RAM operating under the Windows 7 SP1 64 bit operating system.

4.2. Comparison results between MOEA-PUL and the baselines

In this section, we compare the proposed MOEA-PUL with four existing PU learning algorithms, and the results are reported as follows. Table 3 lists the average accuracy values obtained by uPU, PNU, PULD, CL and the proposed MOEA-PUL, where the best result on each dataset is marked in boldface. From the table, it is obvious that MOEA-PUL significantly outperforms the compared algorithms in terms of average accuracy and achieves the best performance on most of the datasets. Specifically, with the increase of μ , MOEA-PUL shows better and more stable performance than the baselines on almost all of datasets. The reason is that the suggested algorithm is not sensitive to the distribution of unlabeled samples and not influenced by the prior information over classes. It can be observed that in low-dimensional datasets, such as *Sonar*, *Heart*, *Ionosphere*, different PU learning algorithms have shown different superiority compared with MOEA-PUL. While for the datasets with larger samples, MOEA-PUL demonstrates the much better performance than the other PU learning algorithms, due to the proposed evolutionary strategies in MOEA-PUL. The win/tie/loss statistics also show that MOEA-PUL is the superiority over all the algorithms, which further indicates that MOEA-PUL is an effective method in solving PU learning problem.

Fig. 5 plots the non-dominated solutions in the space of true positive rate (TPR) and false positive rate (FPR) obtained by MOEA-PUL. The reason of the other four compared algorithms (uPU, PNU, PULD and CL) only having one solution lies in the fact that they are all single-objective based PU learning algorithms and only one solution can be achieved. It can be found that the achieved non-dominated solutions have better trade-off between TPR and FPR than the solutions obtained by the four algorithms for PU learning on all datasets, which demonstrates the superior performance of the proposed algorithm.

As can be seen from Table 3 and Fig. 5, the proposed MOEA-PUL can obtain better performance than the state-of-the-art PU learning algorithms. It should be noted that different from the algorithms that obtain the PU classifiers by learning a noisy label problem (e.g., CL [21]), or make the estimation for the prior information over classes (e.g., uPU [12]) or the assumption for the sample distribution of unlabeled samples (e.g., PULD [11]), the superiority is achieved without any prior knowledge on unlabeled data. To further verify this fact, in the following experiment, we focus on the accuracy of label recognition for unlabeled samples. Table 4 lists the accuracy of label recognition on unlabeled data. Since the comparison results on the experimental datasets with different values of μ are similar, we only show the result in case of $\mu = 0.3$. As shown in this table, the proposed MOEA-PUL can outperform uPU, PNU, PULD and CL on most of the datasets. The win/tie/loss counts show that MOEA-PUL is significantly superior over uPU on all datasets, and better than PNU on ten datasets,

Table 3

The comparison results on the 12 datasets. The best result for each dataset is marked in boldface.

Dataset	μ	uPU	PNU	PULD	CL	MOEA-PUL
Sonar	0.2	0.563 $\pm$ 0.092	0.624 $\pm$ 0.082	0.705 $\pm$ 0.052	0.695 $\pm$ 0.043	0.701 $\pm$ 0.064
	0.3	0.577 $\pm$ 0.093	0.666 $\pm$ 0.088	0.671 $\pm$ 0.035	0.677 $\pm$ 0.046	0.717 $\pm$ 0.075
	0.4	0.551 $\pm$ 0.091	0.683 $\pm$ 0.071	0.676 $\pm$ 0.046	0.658 $\pm$ 0.048	0.712 $\pm$ 0.070
Heart	0.2	0.794 $\pm$ 0.048	0.741 $\pm$ 0.069	0.830 $\pm$ 0.040	0.765 $\pm$ 0.041	0.810 $\pm$ 0.056
	0.3	0.779 $\pm$ 0.058	0.789 $\pm$ 0.064	0.829 $\pm$ 0.033	0.750 $\pm$ 0.043	0.825 $\pm$ 0.043
	0.4	0.780 $\pm$ 0.061	0.806 $\pm$ 0.050	0.830 $\pm$ 0.033	0.739 $\pm$ 0.050	0.827 $\pm$ 0.051
Ionosphere	0.2	0.797 $\pm$ 0.103	0.785 $\pm$ 0.052	0.811 $\pm$ 0.012	0.841 $\pm$ 0.042	0.833 $\pm$ 0.042
	0.3	0.835 $\pm$ 0.068	0.810 $\pm$ 0.058	0.783 $\pm$ 0.053	0.839 $\pm$ 0.043	0.834 $\pm$ 0.041
	0.4	0.811 $\pm$ 0.092	0.837 $\pm$ 0.054	0.732 $\pm$ 0.028	0.785 $\pm$ 0.070	0.830 $\pm$ 0.047
Wdbc	0.2	0.909 $\pm$ 0.071	0.841 $\pm$ 0.038	0.889 $\pm$ 0.062	0.920 $\pm$ 0.028	0.963 $\pm$ 0.022
	0.3	0.905 $\pm$ 0.098	0.869 $\pm$ 0.037	0.870 $\pm$ 0.022	0.901 $\pm$ 0.029	0.959 $\pm$ 0.019
	0.4	0.908 $\pm$ 0.087	0.914 $\pm$ 0.030	0.895 $\pm$ 0.045	0.881 $\pm$ 0.034	0.949 $\pm$ 0.022
Balance	0.2	0.928 $\pm$ 0.028	0.753 $\pm$ 0.045	0.886 $\pm$ 0.035	0.948 $\pm$ 0.016	0.955 $\pm$ 0.017
	0.3	0.921 $\pm$ 0.030	0.860 $\pm$ 0.038	0.872 $\pm$ 0.022	0.926 $\pm$ 0.033	0.952 $\pm$ 0.016
	0.4	0.913 $\pm$ 0.036	0.912 $\pm$ 0.028	0.834 $\pm$ 0.035	0.868 $\pm$ 0.036	0.952 $\pm$ 0.018
Breast	0.2	0.930 $\pm$ 0.027	0.753 $\pm$ 0.043	0.711 $\pm$ 0.069	0.950 $\pm$ 0.021	0.971 $\pm$ 0.016
	0.3	0.921 $\pm$ 0.030	0.855 $\pm$ 0.045	0.955 $\pm$ 0.017	0.943 $\pm$ 0.023	0.973 $\pm$ 0.014
	0.4	0.908 $\pm$ 0.036	0.911 $\pm$ 0.027	0.961 $\pm$ 0.013	0.929 $\pm$ 0.025	0.967 $\pm$ 0.013
Transfusion	0.2	0.684 $\pm$ 0.095	0.769 $\pm$ 0.035	0.768 $\pm$ 0.022	0.766 $\pm$ 0.022	0.799 $\pm$ 0.032
	0.3	0.650 $\pm$ 0.100	0.769 $\pm$ 0.030	0.769 $\pm$ 0.024	0.763 $\pm$ 0.026	0.801 $\pm$ 0.025
	0.4	0.636 $\pm$ 0.121	0.767 $\pm$ 0.031	0.765 $\pm$ 0.026	0.762 $\pm$ 0.070	0.802 $\pm$ 0.030
Pima	0.2	0.686 $\pm$ 0.053	0.723 $\pm$ 0.031	0.741 $\pm$ 0.022	0.740 $\pm$ 0.022	0.786 $\pm$ 0.023
	0.3	0.706 $\pm$ 0.051	0.746 $\pm$ 0.039	0.742 $\pm$ 0.024	0.734 $\pm$ 0.031	0.787 $\pm$ 0.023
	0.4	0.706 $\pm$ 0.047	0.756 $\pm$ 0.038	0.735 $\pm$ 0.024	0.722 $\pm$ 0.037	0.780 $\pm$ 0.028
Diabetes	0.2	0.729 $\pm$ 0.046	0.651 $\pm$ 0.039	0.751 $\pm$ 0.032	0.751 $\pm$ 0.027	0.786 $\pm$ 0.025
	0.3	0.748 $\pm$ 0.036	0.737 $\pm$ 0.044	0.744 $\pm$ 0.052	0.731 $\pm$ 0.029	0.786 $\pm$ 0.021
	0.4	0.751 $\pm$ 0.037	0.758 $\pm$ 0.039	0.735 $\pm$ 0.059	0.671 $\pm$ 0.027	0.782 $\pm$ 0.025
Vehicle	0.2	0.677 $\pm$ 0.051	0.782 $\pm$ 0.031	0.755 $\pm$ 0.021	0.769 $\pm$ 0.025	0.798 $\pm$ 0.022
	0.3	0.678 $\pm$ 0.047	0.785 $\pm$ 0.033	0.752 $\pm$ 0.037	0.766 $\pm$ 0.022	0.799 $\pm$ 0.031
	0.4	0.675 $\pm$ 0.052	0.767 $\pm$ 0.037	0.740 $\pm$ 0.046	0.763 $\pm$ 0.025	0.798 $\pm$ 0.027
German	0.2	0.513 $\pm$ 0.056	0.739 $\pm$ 0.032	0.766 $\pm$ 0.011	0.745 $\pm$ 0.023	0.779 $\pm$ 0.030
	0.3	0.507 $\pm$ 0.084	0.753 $\pm$ 0.032	0.750 $\pm$ 0.016	0.724 $\pm$ 0.028	0.777 $\pm$ 0.028
	0.4	0.496 $\pm$ 0.072	0.740 $\pm$ 0.034	0.732 $\pm$ 0.031	0.739 $\pm$ 0.022	0.769 $\pm$ 0.023
Usps	0.2	0.929 $\pm$ 0.032	0.911 $\pm$ 0.007	0.989 $\pm$ 0.004	0.986 $\pm$ 0.002	0.985 $\pm$ 0.003
	0.3	0.943 $\pm$ 0.013	0.955 $\pm$ 0.005	0.988 $\pm$ 0.003	0.983 $\pm$ 0.003	0.982 $\pm$ 0.002
	0.4	0.954 $\pm$ 0.013	0.976 $\pm$ 0.004	0.988 $\pm$ 0.003	0.977 $\pm$ 0.003	0.975 $\pm$ 0.006
win/tie/loss	0.2	11/1/0	12/0/0	9/1/2	9/3/0	-/-/-
	0.3	11/1/0	11/1/0	10/1/1	10/2/0	-/-/-
	0.4	11/1/0	11/1/0	10/1/1	11/1/0	-/-/-

while for PULD, MOEA-PUL wins for seven out of twelve datasets, ties for four and loses only for once. As for CL, it wins MOEA-PUL on only four datasets. These experimental results further demonstrate the effectiveness of the MOEA-PUL in the case of no any prior knowledge available.

To further show the effectiveness of the proposed MOEA-PUL, a widely used metric in machine learning scenario, F-measure [30], is adopted to evaluate the performance of the compared algorithms, which is also used in PU learning [31]. Table 5 gives the average F-measure values obtained by uPU, PNU, PULD, CL and the suggested MOEA-PUL, where the best result on each dataset is marked in boldface. From Table 5, we can find that MOEA-PUL outperforms the compared algorithms on most of the datasets in terms of F-measure. The win/tie/loss counts show that MOEA-PUL wins or ties for most of datasets and nearly never for loss. This statistics further demonstrate the superiority of the proposed MOEA-PUL.

4.3. Effectiveness of the proposed strategies in MOEA-PUL

As mentioned in Section 3, there are two important strategies suggested in MOEA-PUL. In this section, we empirically investigate the influence of the two strategies on the performance of MOEA-PUL.

In Section 3.3, a PU similarity based initialization strategy is proposed to replace the random initialization method, which

can obtain the initial population with diversity and result in better performance for MOEA-PUL. To verify this fact, we compare MOEA-PUL with MOEA-PUL-Random algorithm, where MOEA-PUL-Random is the same as MOEA-PUL, except that the population is randomly generated. Fig. 6 plots a population generated by MOEA-PUL and a randomly generated population using MOEA-PUL-Random on objective space, where each population contains 100 individuals. From the figure, it can be observed that the population generated by MOEA-PUL has better diversity than the randomly generated population. Table 6 shows the results on twelve datasets. From this table, we can find that due to the obtained initial population with diversity, MOEA-PUL outperforms MOEA-PUL-Random on all datasets in terms of average classification accuracy. The statistical results further demonstrate the effectiveness of the proposed PU similarity based initialization strategy.

To further enhance the performance of MOEA-PUL, an elite label based learning strategy is suggested in Section 3.4. Table 7 shows the results obtained by MOEA-PUL and MOEA-PUL-NonElite, where the proposed elite label based learning strategy is not adopted in MOEA-PUL-NonElite. As shown in this table, MOEA-PUL wins or ties on all datasets and never loses for once, which validates the effectiveness of the proposed elite label based learning strategy.

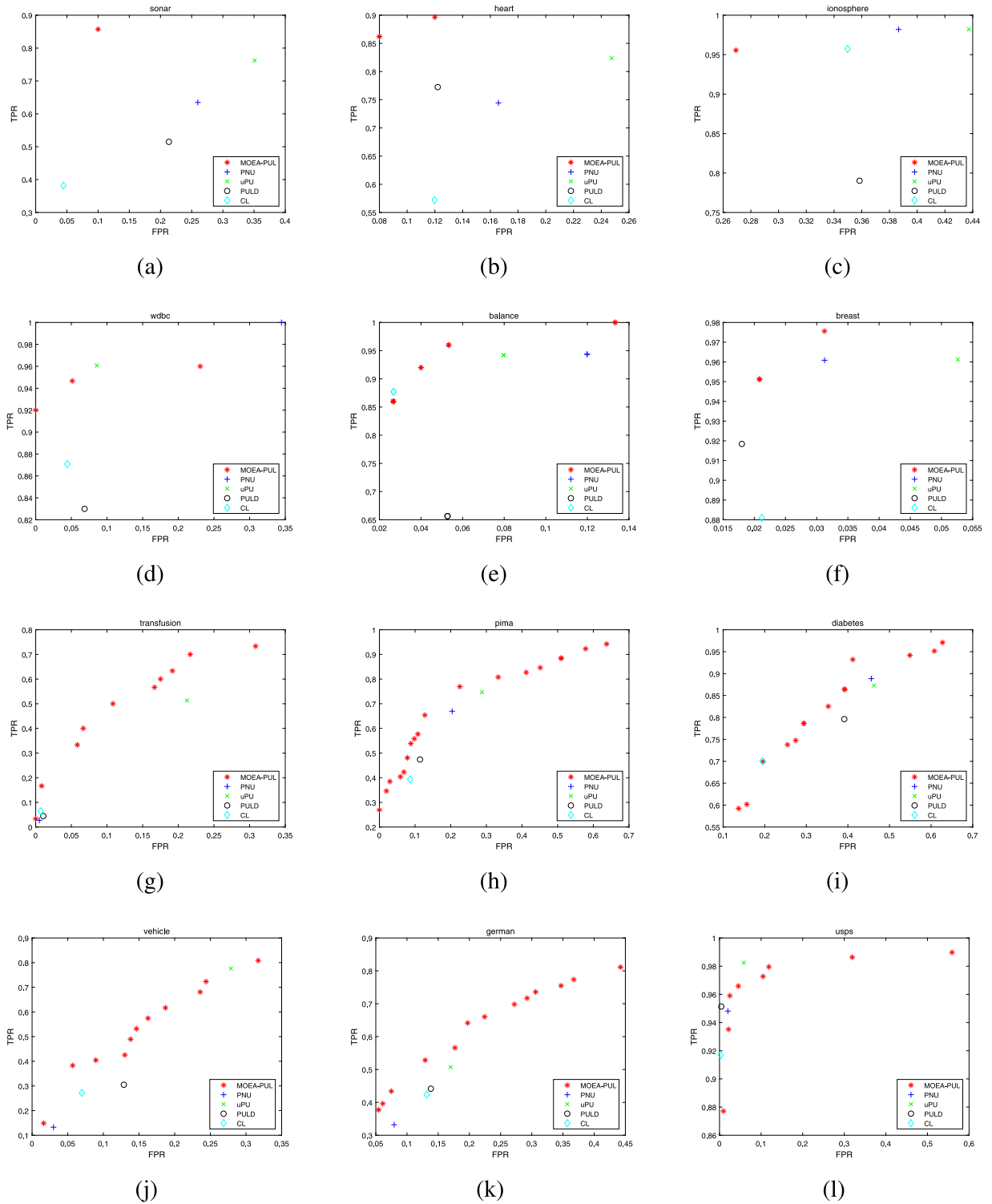


Fig. 5. The final solutions obtained by different PU learning algorithms, where $\mu = 0.3$ in MOEA-PUL.

4.4. Sensitivity analysis of parameters G in MOEA-PUL

As mentioned in Section 3.2, there is one parameter G in the proposed MOEA-PUL for determining when to start the proposed elite label based learning strategy. In this section, we empirically investigate the influence of parameter G on the performance of MOEA-PUL.

Fig. 7 presents the results of the proposed MOEA-PUL on 12 datasets in terms of accuracy, where G increases from 30 to 60 with the interval of 5. As can be seen from the figure, with different values of the parameter G , MOEA-PUL shows different performance on the twelve datasets. It can be observed that when the value of G is set small (such as $G = 45$), the performance of proposed method is not good and the reason is attributed to the fact that the smaller value of G means the fewer elite labels

Table 4

The average accuracy of label recognition for unlabeled samples on the 12 datasets obtained by the comparison algorithms. The best result for each dataset is marked in boldface.

Dataset	uPU	PNU	PULD	CL	MOEA-PUL
Sonar	0.423 ± 0.072	0.681 ± 0.050	0.685 ± 0.025	0.848 ± 0.009	0.738 ± 0.076
Heart	0.698 ± 0.099	0.760 ± 0.054	0.895 ± 0.017	0.894 ± 0.022	0.867 ± 0.013
Ionosphere	0.803 ± 0.042	0.711 ± 0.041	0.748 ± 0.064	0.824 ± 0.019	0.837 ± 0.016
Wdbc	0.938 ± 0.013	0.795 ± 0.026	0.914 ± 0.034	0.951 ± 0.019	0.980 ± 0.008
Balance	0.934 ± 0.027	0.805 ± 0.029	0.918 ± 0.030	0.961 ± 0.007	0.970 ± 0.006
Breast	0.915 ± 0.056	0.808 ± 0.032	0.969 ± 0.004	0.963 ± 0.008	0.973 ± 0.005
Transfusion	0.806 ± 0.094	0.911 ± 0.002	0.913 ± 0.003	0.909 ± 0.006	0.903 ± 0.019
Pima	0.702 ± 0.132	0.763 ± 0.015	0.830 ± 0.011	0.860 ± 0.020	0.789 ± 0.007
Diabetes	0.713 ± 0.032	0.563 ± 0.048	0.728 ± 0.024	0.756 ± 0.016	0.780 ± 0.009
Vehicle	0.652 ± 0.029	0.878 ± 0.012	0.855 ± 0.011	0.875 ± 0.012	0.900 ± 0.004
German	0.453 ± 0.075	0.823 ± 0.017	0.848 ± 0.007	0.855 ± 0.006	0.885 ± 0.008
Usps	0.937 ± 0.028	0.951 ± 0.001	0.992 ± 0.001	0.992 ± 0.001	0.991 ± 0.001
win/tie/loss	12/0/0	10/2/0	7/4/1	6/3/3	

Table 5

The average F-measure values on the 12 datasets. The best result for each dataset is marked in boldface.

Dataset	μ	uPU	PNU	PULD	CL	MOEA-PUL
Sonar	0.2	0.688 ± 0.053	0.698 ± 0.084	0.604 ± 0.103	0.630 ± 0.074	0.704 ± 0.121
	0.3	0.661 ± 0.066	0.699 ± 0.038	0.584 ± 0.067	0.511 ± 0.084	0.640 ± 0.102
	0.4	0.655 ± 0.115	0.582 ± 0.070	0.583 ± 0.057	0.491 ± 0.114	0.585 ± 0.226
Heart	0.2	0.782 ± 0.034	0.739 ± 0.055	0.801 ± 0.057	0.694 ± 0.069	0.770 ± 0.060
	0.3	0.739 ± 0.075	0.769 ± 0.039	0.788 ± 0.046	0.725 ± 0.054	0.811 ± 0.070
	0.4	0.769 ± 0.078	0.807 ± 0.039	0.802 ± 0.052	0.621 ± 0.105	0.812 ± 0.042
Ionosphere	0.2	0.764 ± 0.021	0.847 ± 0.031	0.870 ± 0.018	0.891 ± 0.028	0.911 ± 0.020
	0.3	0.785 ± 0.065	0.861 ± 0.063	0.835 ± 0.049	0.875 ± 0.051	0.897 ± 0.034
	0.4	0.898 ± 0.039	0.904 ± 0.018	0.795 ± 0.029	0.796 ± 0.086	0.858 ± 0.043
Wdbc	0.2	0.958 ± 0.032	0.880 ± 0.043	0.904 ± 0.063	0.934 ± 0.012	0.968 ± 0.017
	0.3	0.940 ± 0.036	0.905 ± 0.018	0.887 ± 0.029	0.916 ± 0.028	0.964 ± 0.011
	0.4	0.955 ± 0.016	0.936 ± 0.022	0.909 ± 0.046	0.889 ± 0.017	0.968 ± 0.016
Balance	0.2	0.926 ± 0.035	0.783 ± 0.039	0.825 ± 0.129	0.941 ± 0.016	0.914 ± 0.018
	0.3	0.925 ± 0.021	0.873 ± 0.029	0.782 ± 0.178	0.904 ± 0.036	0.906 ± 0.019
	0.4	0.915 ± 0.036	0.886 ± 0.024	0.746 ± 0.178	0.835 ± 0.026	0.924 ± 0.015
Breast	0.2	0.911 ± 0.035	0.780 ± 0.062	0.405 ± 0.118	0.930 ± 0.021	0.960 ± 0.019
	0.3	0.893 ± 0.037	0.868 ± 0.053	0.931 ± 0.033	0.917 ± 0.025	0.958 ± 0.016
	0.4	0.891 ± 0.032	0.921 ± 0.027	0.940 ± 0.023	0.891 ± 0.037	0.955 ± 0.018
Transfusion	0.2	0.411 ± 0.210	0.243 ± 0.061	0.102 ± 0.037	0.181 ± 0.107	0.546 ± 0.061
	0.3	0.487 ± 0.047	0.109 ± 0.073	0.121 ± 0.011	0.124 ± 0.066	0.533 ± 0.067
	0.4	0.448 ± 0.043	0.130 ± 0.105	0.102 ± 0.031	0.114 ± 0.083	0.513 ± 0.070
Pima	0.2	0.651 ± 0.054	0.663 ± 0.056	0.568 ± 0.051	0.550 ± 0.055	0.685 ± 0.025
	0.3	0.647 ± 0.032	0.653 ± 0.057	0.561 ± 0.056	0.527 ± 0.060	0.678 ± 0.058
	0.4	0.668 ± 0.033	0.629 ± 0.068	0.581 ± 0.037	0.440 ± 0.083	0.679 ± 0.025
Diabetes	0.2	0.801 ± 0.053	0.789 ± 0.023	0.813 ± 0.028	0.793 ± 0.030	0.844 ± 0.030
	0.3	0.822 ± 0.033	0.822 ± 0.019	0.801 ± 0.054	0.768 ± 0.033	0.848 ± 0.024
	0.4	0.815 ± 0.017	0.814 ± 0.020	0.793 ± 0.057	0.706 ± 0.023	0.842 ± 0.032
Vehicle	0.2	0.577 ± 0.016	0.570 ± 0.043	0.291 ± 0.100	0.450 ± 0.050	0.632 ± 0.033
	0.3	0.611 ± 0.053	0.420 ± 0.041	0.378 ± 0.046	0.392 ± 0.096	0.620 ± 0.072
	0.4	0.587 ± 0.055	0.208 ± 0.095	0.358 ± 0.114	0.368 ± 0.107	0.614 ± 0.081
German	0.2	0.546 ± 0.032	0.585 ± 0.048	0.529 ± 0.041	0.532 ± 0.039	0.639 ± 0.027
	0.3	0.535 ± 0.046	0.520 ± 0.094	0.497 ± 0.055	0.496 ± 0.046	0.611 ± 0.092
	0.4	0.530 ± 0.047	0.434 ± 0.075	0.496 ± 0.060	0.467 ± 0.043	0.622 ± 0.048
Usps	0.2	0.800 ± 0.028	0.787 ± 0.015	0.967 ± 0.009	0.957 ± 0.005	0.948 ± 0.008
	0.3	0.862 ± 0.038	0.879 ± 0.006	0.964 ± 0.007	0.939 ± 0.007	0.943 ± 0.007
	0.4	0.871 ± 0.038	0.926 ± 0.012	0.964 ± 0.007	0.929 ± 0.012	0.930 ± 0.015
win/tie/loss	0.2	6/6/0	8/4/0	6/5/1	6/5/1	
	0.3	4/8/0	6/6/0	6/5/1	7/5/0	
	0.4	4/8/0	6/6/0	6/5/1	9/3/0	

obtained. Meanwhile, at the later stages of evolution (such as $G > 50$), the effectiveness of the guidance of the elite label is weakened, since the difference between the elite individuals and the other individuals is little. In summary, MOEA-PUL can obtain better performance on most of the datasets, when G is set to 50.

5. Conclusions and future work

In this paper, we have proposed an MOEA without using any prior knowledge for PU learning problem. To be specific, a new metric named UAR is firstly designed to measure the quality of a set of candidate labels for unlabeled data, which is combined with

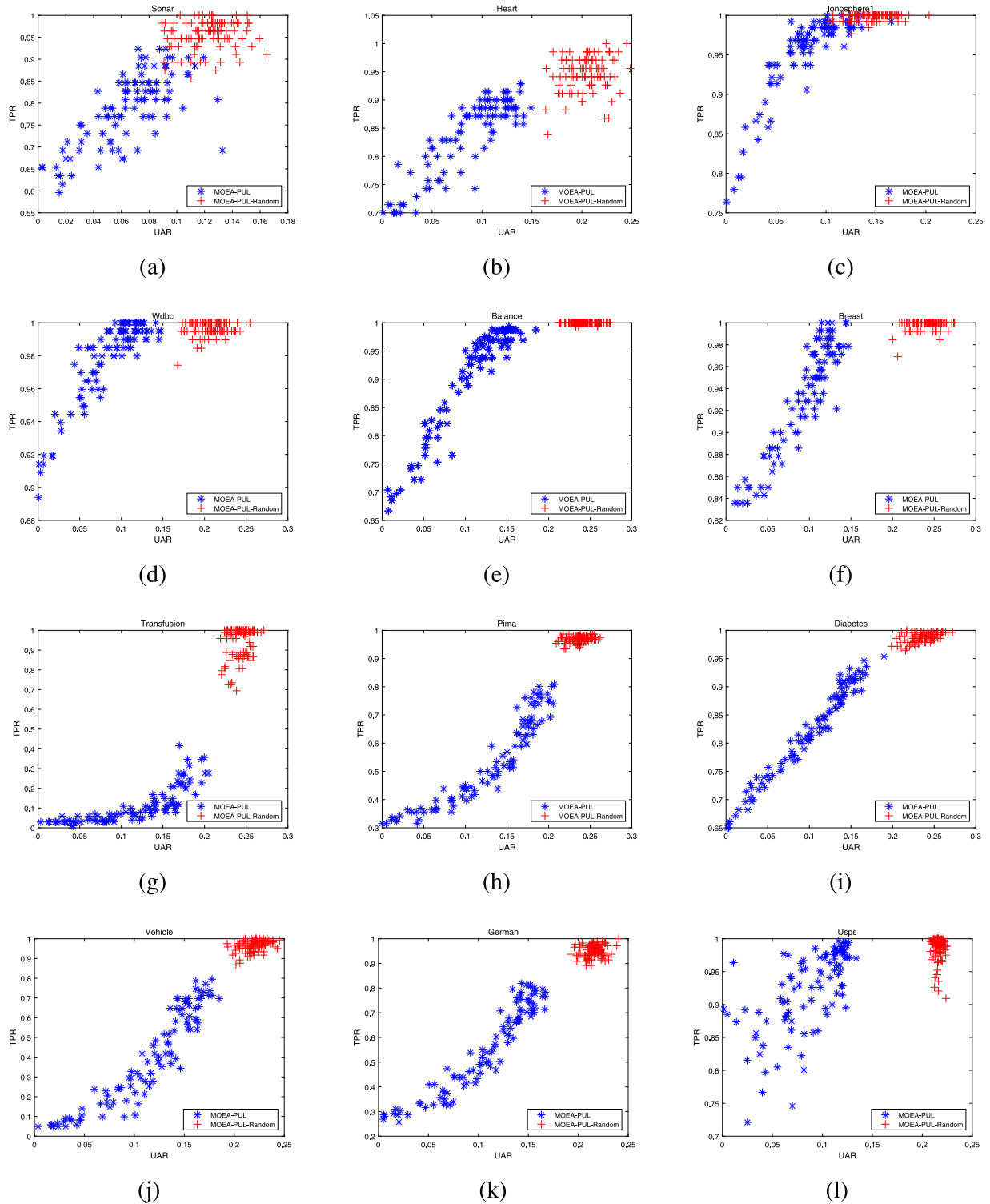


Fig. 6. An illustration of the effectiveness of the proposed initialization strategy compared with the random initialization on objective space.

TPR and formulate the PU learning into a bi-objective optimization problem. Then an MOEA, termed MOEA-PUL, is suggested to solve the problem. To further improve the performance of the proposed algorithm, a PU similarity based population initialization strategy and an elite label based learning strategy are developed. Experimental results on different datasets have demonstrated the superiority of the proposed algorithm.

The results of this work have shown that MOEA is a promising method for solving the PU learning problem, which can achieve

PU classifier without using any prior knowledge. In the future, we continue this research line to obtain the PU classifier with high quality. For example, as the first MOEA for PU learning, the proposed algorithm adopt classical NSGA-II as the framework, however, it is well known that in the literatures of MOEA, there also exist other effective frameworks, such as MOEA/D [32], SPEA2 [33], KnEA [34], etc. We plan to design more competitive MOEAs for PU learning. Furthermore, in the proposed MOEA-PUL, we only utilize the unlabeled data for PU learning. However, since

Table 6

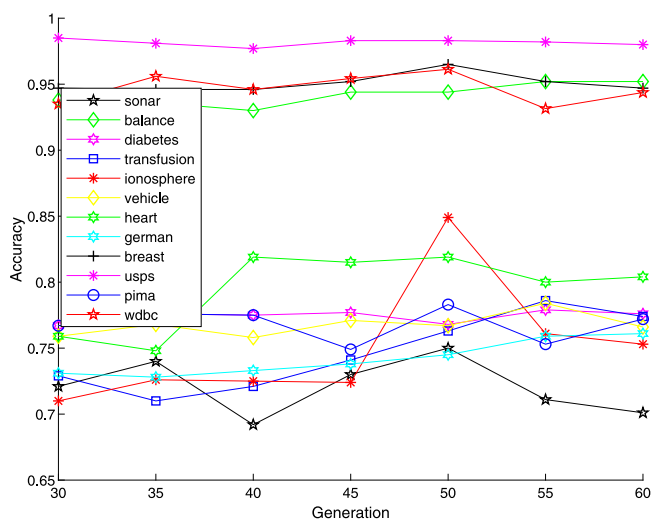
The comparison results between MOEA-PUL and MOEA-PUL-Random on the 12 datasets. The best result for each dataset is marked in boldface.

Dataset	μ	MOEA-PUL-Random	MOEA-PUL
Sonar	0.2	0.553 $\pm$ 0.050	0.701 $\pm$ 0.064
	0.3	0.581 $\pm$ 0.048	0.717 $\pm$ 0.075
	0.4	0.562 $\pm$ 0.075	0.712 $\pm$ 0.070
Heart	0.2	0.674 $\pm$ 0.055	0.810 $\pm$ 0.056
	0.3	0.619 $\pm$ 0.048	0.825 $\pm$ 0.043
	0.4	0.652 $\pm$ 0.042	0.827 $\pm$ 0.051
Ionosphere	0.2	0.792 $\pm$ 0.031	0.833 $\pm$ 0.042
	0.3	0.798 $\pm$ 0.021	0.834 $\pm$ 0.041
	0.4	0.798 $\pm$ 0.048	0.830 $\pm$ 0.047
Wdbc	0.2	0.826 $\pm$ 0.066	0.963 $\pm$ 0.022
	0.3	0.856 $\pm$ 0.035	0.959 $\pm$ 0.019
	0.4	0.894 $\pm$ 0.042	0.949 $\pm$ 0.022
Balance	0.2	0.762 $\pm$ 0.030	0.955 $\pm$ 0.017
	0.3	0.782 $\pm$ 0.023	0.952 $\pm$ 0.016
	0.4	0.819 $\pm$ 0.048	0.952 $\pm$ 0.018
Breast	0.2	0.755 $\pm$ 0.033	0.971 $\pm$ 0.016
	0.3	0.772 $\pm$ 0.043	0.973 $\pm$ 0.014
	0.4	0.772 $\pm$ 0.039	0.967 $\pm$ 0.013
Transfusion	0.2	0.253 $\pm$ 0.056	0.799 $\pm$ 0.032
	0.3	0.250 $\pm$ 0.050	0.801 $\pm$ 0.025
	0.4	0.249 $\pm$ 0.027	0.802 $\pm$ 0.030
Pima	0.2	0.533 $\pm$ 0.045	0.786 $\pm$ 0.023
	0.3	0.544 $\pm$ 0.039	0.787 $\pm$ 0.023
	0.4	0.557 $\pm$ 0.061	0.780 $\pm$ 0.028
Diabetes	0.2	0.706 $\pm$ 0.027	0.776 $\pm$ 0.022
	0.3	0.707 $\pm$ 0.026	0.794 $\pm$ 0.039
	0.4	0.708 $\pm$ 0.035	0.773 $\pm$ 0.024
Vehicle	0.2	0.565 $\pm$ 0.023	0.786 $\pm$ 0.025
	0.3	0.563 $\pm$ 0.019	0.786 $\pm$ 0.021
	0.4	0.593 $\pm$ 0.019	0.782 $\pm$ 0.025
German	0.2	0.503 $\pm$ 0.027	0.779 $\pm$ 0.030
	0.3	0.495 $\pm$ 0.021	0.777 $\pm$ 0.028
	0.4	0.530 $\pm$ 0.025	0.769 $\pm$ 0.023
Usps	0.2	0.556 $\pm$ 0.174	0.983 $\pm$ 0.002
	0.3	0.459 $\pm$ 0.065	0.982 $\pm$ 0.005
	0.4	0.534 $\pm$ 0.117	0.975 $\pm$ 0.006
win/tie/loss	0.2	12/0/0	-/-/-
	0.3	12/0/0	-/-/-
	0.4	11/1/0	-/-/-

Table 7

The comparison results between MOEA-PUL and MOEA-PUL-NonElite on the 12 datasets. The best result for each dataset is marked in boldface.

Dataset	μ	MOEA-PUL-NonElite	MOEA-PUL
Sonar	0.2	0.702 $\pm$ 0.045	0.701 $\pm$ 0.064
	0.3	0.688 $\pm$ 0.065	0.717 $\pm$ 0.075
	0.4	0.702 $\pm$ 0.044	0.712 $\pm$ 0.070
Heart	0.2	0.815 $\pm$ 0.057	0.810 $\pm$ 0.056
	0.3	0.815 $\pm$ 0.029	0.825 $\pm$ 0.043
	0.4	0.796 $\pm$ 0.019	0.827 $\pm$ 0.051
Ionosphere	0.2	0.846 $\pm$ 0.040	0.833 $\pm$ 0.042
	0.3	0.792 $\pm$ 0.053	0.834 $\pm$ 0.041
	0.4	0.812 $\pm$ 0.045	0.830 $\pm$ 0.047
Wdbc	0.2	0.968 $\pm$ 0.013	0.963 $\pm$ 0.022
	0.3	0.973 $\pm$ 0.014	0.959 $\pm$ 0.019
	0.4	0.954 $\pm$ 0.027	0.949 $\pm$ 0.022
Balance	0.2	0.950 $\pm$ 0.021	0.955 $\pm$ 0.017
	0.3	0.947 $\pm$ 0.012	0.952 $\pm$ 0.016
	0.4	0.939 $\pm$ 0.029	0.952 $\pm$ 0.018
Breast	0.2	0.958 $\pm$ 0.033	0.971 $\pm$ 0.016
	0.3	0.772 $\pm$ 0.043	0.973 $\pm$ 0.014
	0.4	0.772 $\pm$ 0.039	0.967 $\pm$ 0.013
Transfusion	0.2	0.785 $\pm$ 0.027	0.799 $\pm$ 0.032
	0.3	0.779 $\pm$ 0.032	0.801 $\pm$ 0.025
	0.4	0.787 $\pm$ 0.043	0.802 $\pm$ 0.030
Pima	0.2	0.782 $\pm$ 0.038	0.786 $\pm$ 0.023
	0.3	0.785 $\pm$ 0.031	0.787 $\pm$ 0.023
	0.4	0.776 $\pm$ 0.097	0.780 $\pm$ 0.028
Diabetes	0.2	0.747 $\pm$ 0.036	0.786 $\pm$ 0.025
	0.3	0.758 $\pm$ 0.012	0.786 $\pm$ 0.021
	0.4	0.763 $\pm$ 0.034	0.782 $\pm$ 0.025
Vehicle	0.2	0.780 $\pm$ 0.012	0.798 $\pm$ 0.022
	0.3	0.780 $\pm$ 0.021	0.799 $\pm$ 0.031
	0.4	0.781 $\pm$ 0.031	0.798 $\pm$ 0.027
German	0.2	0.762 $\pm$ 0.027	0.779 $\pm$ 0.030
	0.3	0.747 $\pm$ 0.017	0.777 $\pm$ 0.028
	0.4	0.758 $\pm$ 0.050	0.769 $\pm$ 0.023
Usps	0.2	0.984 $\pm$ 0.004	0.985 $\pm$ 0.003
	0.3	0.982 $\pm$ 0.003	0.982 $\pm$ 0.002
	0.4	0.978 $\pm$ 0.004	0.975 $\pm$ 0.006
win/tie/loss	0.2	3/9/0	-/-/-
	0.3	5/7/0	-/-/-
	0.4	3/9/0	-/-/-

**Fig. 7.** The accuracy of MOEA-PUL on 12 datasets with varying G.

some positive samples in PU data may be contaminated in real applications, as one of the future works, we would like to propose

an effective strategy to select the sample subset from contaminated positive samples for further enhancing the performance of PU classifier.

CRedit authorship contribution statement

Jianfeng Qiu: Methodology, Software, Original draft preparation. **Xiaoqiang Cai:** Data curation, Experiments. **Xingyi Zhang:** Visualization, Investigation, Reviewing. **Fan Cheng:** Supervision, Conceptualization. **Shenzhi Yuan:** Software, Validation. **Guanglong Fu:** Writing and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (Grant No. U1804262, 61976001, 61876184 and 62076001), Humanities and Social Sciences Project of Chinese

Ministry of Education (Grant No. 18YJC870004), the Natural Science Foundation of Anhui Province, China (Grant No. 1708085 MF166, 1908085MF219 and 2008085QF308), the Anhui Provincial Key Research and Development Plan, China (Grand No. 202004j07 020005) and the Opening Project of Jiangsu Key Laboratory of Data Science and Smart Software (Grant No. 2019DS304)

References

- [1] E. Sansone, F.G. De Natale, Z.H. Zhou, Efficient training for positive unlabeled learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 41 (11) (2018) 2584–2598.
- [2] H. Wei, M. Li, Positive and unlabeled learning for detecting software functional clones with adversarial training, in: *International Joint Conference on Artificial Intelligence*, 2018, pp. 2840–2846.
- [3] X. Li, B. Liu, Learning to classify texts using positive and unlabeled data, in: *International Joint Conference on Artificial Intelligence*, Vol. 3, 2003, pp. 587–592.
- [4] B. Liu, W.S. Lee, P.S. Yu, X. Li, Partially supervised classification of text documents, in: *International Conference on Machine Learning*, Vol. 2, 2002, pp. 387–394.
- [5] H. Yu, J. Han, K.C. Chuan, PEBL: positive example based learning for web page classification using SVM, in: *International Conference on Knowledge Discovery and Data Mining*, 2002, pp. 239–248.
- [6] H. Yu, Single-class classification with mapping convergence, *Mach. Learn.* 61 (1–3) (2005) 49–69.
- [7] W.S. Lee, B. Liu, Learning with positive and unlabeled examples using weighted logistic regression, in: *International Conference on Machine Learning*, Vol. 3, 2003, pp. 448–455.
- [8] C. Elkan, K. Noto, Learning classifiers from only positive and unlabeled data, in: *International Conference on Knowledge Discovery and Data Mining*, 2008, pp. 213–220.
- [9] C. Gong, T. Liu, J. Yang, D. Tao, Large-margin label-calibrated support vector machines for positive and unlabeled learning, *IEEE Trans. Neural Netw. Learn. Syst.* 30 (11) (2019) 3471–3483.
- [10] M.C. Du Plessis, G. Niu, M. Sugiyama, Convex formulation for learning from positive and unlabeled data, in: *International Conference on Machine Learning*, 2015, pp. 1386–1394.
- [11] C. Zhang, D. Ren, T. Liu, J. Yang, C. Gong, Positive and unlabeled learning with label disambiguation, in: *International Joint Conference on Artificial Intelligence*, 2019, pp. 4250–4256.
- [12] M.C. Du Plessis, G. Niu, M. Sugiyama, Analysis of learning from positive and unlabeled data, in: *Advances in Neural Information Processing Systems*, 2014, pp. 703–711.
- [13] T. Gong, G. Wang, J. Ye, Z. Xu, M. Lin, Margin based PU learning, in: *AAAI Conference on Artificial Intelligence*, 2018, pp. 1–8.
- [14] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multiobjective genetic algorithm: NSGA-II, *IEEE Trans. Evol. Comput.* 6 (2) (2002) 182–197.
- [15] K. Deb, H. Jain, An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part I: Solving problems with box constraints, *IEEE Trans. Evol. Comput.* 18 (4) (2014) 577–601.
- [16] H. Seada, M. Abouhawwash, K. Deb, Multiphase balance of diversity and convergence in multiobjective optimization, *IEEE Trans. Evol. Comput.* 23 (3) (2019) 503–513.
- [17] J. Qiu, M. Liu, L. Zhang, W. Li, F. Cheng, A multi-level knee point based multi-objective evolutionary algorithm for AUC maximization, *Memet. Comput.* 11 (3) (2019) 285–296.
- [18] W. Hong, K. Tang, Convex hull-based multi-objective evolutionary computation for maximizing receiver operating characteristics performance, *Memet. Comput.* 8 (1) (2016) 35–44.
- [19] Y. Xiao, B. Liu, J. Yin, L. Cao, C. Zhang, Z. Hao, Similarity-based approach for positive and unlabeled learning, in: *International Joint Conference on Artificial Intelligence*, Vol. 22, 2011, pp. 1577–1582.
- [20] H. Shi, S. Pan, J. Yang, C. Gong, Positive and unlabeled learning via loss decomposition and centroid estimation, in: *International Joint Conference on Artificial Intelligence*, 2018, pp. 2689–2695.
- [21] C.G. Northcutt, L. Jiang, L.L. Chuang, Confident learning: Estimating uncertainty in dataset labels, *arXiv:1911.00068v2*.
- [22] K. Nigam, A.K. McCallum, S. Thrun, T. Mitchell, Text classification from labeled and unlabeled documents using EM, *Mach. Learn.* 39 (2–3) (2000) 103–134.
- [23] B. Liu, Y. Dai, X. Li, W.S. Lee, S.Y. Philip, Building text classifiers using positive and unlabeled examples, in: *International Conference on Data Mining*, Vol. 3, 2003, pp. 179–188.
- [24] P. Yang, S.J. Humphrey, D.E. James, Y.H. Yang, R. Jothi, Positive-unlabeled ensemble learning for kinase substrate prediction from dynamic phosphoproteomics data, *Bioinformatics* 32 (2) (2016) 252–259.
- [25] M.C. Du Plessis, M. Sugiyama, Class prior estimation from positive and unlabeled data, *IEEE Trans. Inf. Syst.* 97 (5) (2014) 1358–1362.
- [26] R. Jin, Z. Ghahramani, Learning with multiple labels, in: *Advances in Neural Information Processing Systems*, 2003, pp. 921–928.
- [27] T. Sakai, M.C. du Plessis, G. Niu, M. Sugiyama, Semi-supervised classification based on classification from positive and unlabeled data, in: *International Conference on Machine Learning*, Vol. 70, 2017, pp. 2998–3006.
- [28] F. Cheng, G. Fu, X. Zhang, J. Qiu, Multi-objective evolutionary algorithm for optimizing the partial area under the ROC curve, *Knowl.-Based Syst.* 170 (2019) 61–69.
- [29] C.M. Rocco, J.E. Ramirez-Marquez, D.E. Salazar, C. Yajure, Assessing the vulnerability of a power system through a multiple objective contingency screening approach, *IEEE Trans. Reliab.* 60 (2) (2011) 394–403.
- [30] R. Soleymani, E. Granger, G. Fumera, F-measure curves: A tool to visualize classifier performance under imbalance, *Pattern Recognit.* (2019) 1–19.
- [31] D.H. Fusilier, M. Montes-Y-Gomez, P. Rosso, R.G. Cabrera, Detecting positive and negative deceptive opinions using PU-learning, *Inf. Process. Manage.* 51 (4) (2015) 433–443.
- [32] Q. Zhang, H. Li, MOEA/D: a multiobjective evolutionary algorithm based on decomposition, *IEEE Trans. Evol. Comput.* 11 (6) (2007) 712–731.
- [33] E. Zitzler, M. Laumanns, L. Thiele, SPEA2: Improving the strength pareto evolutionary algorithm, in: *Proceedings of the Fifth Conference on Evolutionary Methods for Design, Optimization and Control with Applications to Industrial Problems*, 2001, pp. 95–100.
- [34] X. Zhang, Y. Tian, Y. Jin, A knee point-driven evolutionary algorithm for many-objective optimization, *IEEE Trans. Evol. Comput.* 19 (6) (2015) 761–776.